

Smart Search Engine Using Artificial Intelligence

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ABSTRACT

Due to the increasing information on the Web it becomes very difficult for a single search engine to provide comprehensive Web coverage, thus increasing the difficulty for an average user to search information on the web. We have proposed an idea for smart Meta search engine which will have a more comprehensive coverage of the web by targeting different search engines for different search categories. The user's search query will be modified using a knowledge base and appropriate search engine will be selected. The results returned from these search engines will be merged and ranked. These results will be classified into various categories using a classification algorithm and then the result will be displayed sorted into categories. It will be beneficial to the user who has an obscure idea about the information he/she wanted to search.

Categories and Subject Descriptors

H.3.3 [Information Search and Retrieval]: Search process

General Terms

Algorithms, Design, Management, Reliability, Performance

Keywords

Meta search engine, artificial intelligence, classification, information retrieval

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ICWET'11, February 25–26, 2011, Mumbai, Maharashtra, India.

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1. INTRODUCTION

The information on the web is overwhelming so searching information on the web becomes very difficult for an average user using traditional search engines. A meta-search engine is a search tool which sends the search query to multiple search engines, as a result the information is more complete and the coverage of the web is improved. A meta search engine itself does not crawl the webpages; it passes the query to the other search engines and fetches the results from them. The results from the various search engines are then merged according to the page merging and page ranking algorithm. Since, the existing Meta search engines are not context based and do not refine the search results according to a particular individual profile, we have proposed a meta search engine which uses

Artificial intelligence to modify the search query and then classify the results into various categories based on context. We have even proposed to use a page ranking algorithm which will adapt itself according to the users profile for a better surfing experience. A summary for the links will also be displayed so that the user can refer the summary and then visit a particular link. This also increases the user friendliness. The paper discusses the existing system like Google, context less Meta search engine and proposed system

2. OVERVIEW OF CURRENT SYSTEM

2.1 Google Search Engine Architectural Working

In Google, the web crawling is done by several distributed crawlers. There is an URLserver that sends lists of URLs to be fetched to the crawlers. The web pages that are fetched are then sent to the storeserver which compresses and stores the web pages into a repository. Every web page has an associated ID number called a docID which is assigned whenever a new URL is parsed out of a web page.

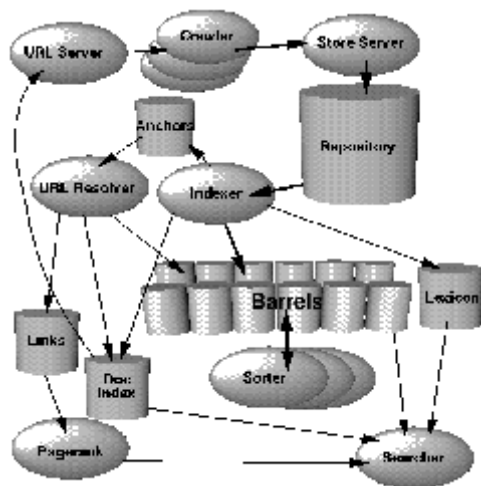


Figure 1: Google search engine architecture

The indexing function is performed by the indexer and the sorter. Each document is converted into a set of word occurrences called hits. The indexer distributes these hits into a set of "barrels", creating a partially sorted forward index. The indexer parses out all the links in every web page and stores important information about them in an anchors file which contains enough information to determine where each link points from and to, and the text of the link. The URLresolver reads the anchors file and converts relative URLs into absolute URLs and in turn into docIDs. It also generates a database of links which are pairs of docIDs. The links database is used to compute PageRanks for all the documents. The sorter takes the barrels and generates the inverted index. The sorter also produces a list of wordIDs and offsets into the inverted index. A program called DumpLexicon takes this list together with the lexicon produced by the indexer and generates a new lexicon to be used by the searcher. The searcher is run by a web server and uses the lexicon built by DumpLexicon together with the inverted index and the PageRanks to answer queries

2.2. Meta Search Engine

The Architecture of the Meta Search Engine is different from traditional search engine since it does not search the web. Meta search engines operate on the assumption that the Web is too large for any one search engine to index it all and that better search results can be obtained by combining the results from several search engines. It passes the query to other search engines and then compiles the result. After compiling the result from other search engines, a Meta Search Engine presents the results to the user. Presently the Meta Search Engines that are existing only use a fix set of search engines to search a particular query, and all the search engines are general purpose search engines i.e. they are used to search the entire web and not on

specific domains. Architecture of Meta Search Engine is given in Figure 2.

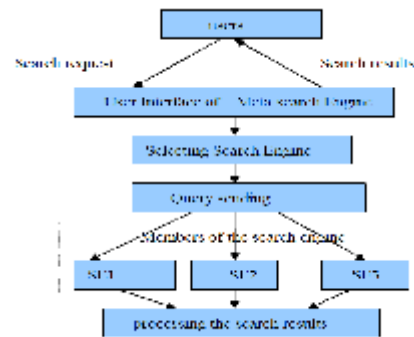


Figure 2: Meta Search Engine Architecture

The user's query will be sent to various search engines. The results will then be extracted from the selected search engines and will be merged according to appropriate page merging algorithm. The merged results will then be ranked. This ranking module will be different for different Meta search engines. These results will then be compiled and presented to the user.

3. PROPOSED ARCHITECTURE

When the user types a query it will go to the query processor. The Query Processor will pass the query to the knowledge base. The knowledge base will return the various categories the particular query might be related to, like if monkey is the query

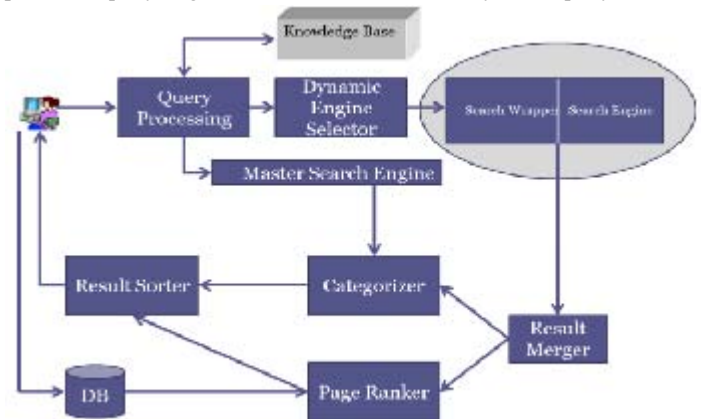


Figure 3: Proposed Architecture

that user wants to search, then knowledge base will return monkey animal, monkey networks and so on. The same query is also passed to the Dynamic Engine Selector which will select particular search engines depending on the type of query as specified in Table 1.

Table 1: List of Search Engines

Type	Search Engine
General Purpose Search	Yahoo
	Bing
	Ask
Files	Torrentz
Mathematics	Wolframalpha
Medical	Novoseek
News and Social Web	Spinn3r
Videos	Blinkz
Educational	SweetSearch

The selected search engines will then process the particular query and the result of all the selected search engines are given to the result merger. The result merger will merge the result and remove all the duplicates. The output of the result merger is given to the page ranker and categorizer.

The master search engine (Google search engine) will get the modified search query from the query processor. The results returned by the master search engine will act as training set for the results returned by the other search engines. The results returned by the other search engines will be classified using Naïve Bayesian Classifier based on the training set. The pages will be ranked according to a page ranking algorithm. After ranking, the results will be sorted in the page sorter which will sort the various pages in appropriate categories depending on the output of the categorizer and the page ranker. This result will then be presented to the user in categorized form. Depending on the user's interaction with the system, the heuristics of the page ranking algorithm will be changed and stored in the database and referenced for future page ranking

4. OUR APPROACH

For the knowledge base we will be using OpenCyc in conjunction with Online Encyclopedia. The Dynamic Engine Selector will select search engines depending on the search query as mentioned in Table 1. The results will then be merged removing the duplicates. This will be done by storing the results as a hash value in the hash tables. If a link returns a hash value which corresponds to the same hash value, then the entry will be considered as duplicate and removed. These merged results will then be passed to the categorizer. For Text categorizing we will be using Naïve Bayesian Classifier Algorithm. For the Classification Algorithm, the training set will be the results returned by the master search engine (Google search engine). The results returned by the other search engines will be classified according to the training set. The page rank algorithm will be used to assign a heuristic page rank to every page. For that purpose we have chosen the following parameters.

Table 2: Page Ranking Parameters

DaysOld	Age of the page. Newer page will be given higher score
WordCount	Number of words on the page
PopularityCount	Number of inlinks to the page
ImageCount	Number of images on the page
NoKeywords	Number of times the keyword appears on the page. The frequency alongwith the level is considered
PathLength	It signifies the depth of the page from the main page. Greater the depth, less relevant the page is.
Summary	The summary of the page will be semantically compared to the query to determine the relevance
Topicalrelevance	How much the page is relevant to the context of search
SourceEngine	Indicates the search engine returning the result. Different engines are given preferences for different search contexts
RankInSource	The rank of the page in the engine it was fetched from

For each parameter a weight age is assigned. This weight age is assigned based on initial statistical analysis. The weight age is then constantly adjusted based on user feedback.

$$\text{PageRank}(P) = \sum P_i(P) * W_i(P_i)$$

$P_i \rightarrow$ Value of the parameter in the page P

$W_i \rightarrow$ Weight of the parameter P_i

For the purpose of normalizing the page rank, it is multiplied by a dampening factor which converts the page rank to a number in the range (0, 1)

$$f(x) = 2 * 11 + e^{-x} - 0.5$$

$$\text{PageRank}(P) = f(x) * \text{PageRank}(P)$$

This gives a heuristic page rank to the page in the range (0, 1)

5. CONCLUSION

In the paper we have studied the existing systems like google and context less Meta Search Engine with their drawbacks and proposed a context based Meta Search Engine resolving the drawbacks and adding new features.

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